Kwang Hak Kim

Ph.D. Candidate, University of California San Diego, La Jolla, CA
kwk001@ucsd.edu — +1 (858) 245-4890 — kwanghakkim.github.io

EDUCATION

Ph.D. Candidate in Mechanical Engineering Sept. 2022 – Jun. 2027 (expected)
University of California San Diego, La Jolla, CA

M.S. in Mechanical Engineering Sept. 2022 – May 2024
University of California San Diego, La Jolla, CA

B.S. in Aerospace Engineering Aug. 2016 – May 2020
The University of Texas at Austin, Austin, TX

RESEARCH EXPERIENCE

Nonlinear and Adaptive Control Laboratory, UC San Diego La Jolla, CA
Graduate Student Researcher Sept. 2022 to Present
Advisors: Prof. Miroslav Krstić and Prof. Mamadou Diagne

- Developed novel **stabilization and safety-critical** control strategies for **nonholonomic underactuated vehicles**
- Designed safety filters for **autonomous traffic management** on aircraft carrier decks, and engaged in **invited technical discussions with NAWCAD** and contributed to **ONR annual reviews**
- Constructed globally strict CLFs for unicycle and Dubins vehicle models, enabling **smooth time-invariant, inverse-optimal, adaptive, and prescribed/fixed-time stabilization**
- Designed novel smooth robust CBFs for high-relative-degree systems to guarantee **collision avoidance with unknown moving obstacles**
- Introduced a constant-sum high-order CBF framework that guarantees **safety between parallel boundaries** by eliminating loss of control authority
- Extending stabilization and safety-critical control frameworks to **broader classes of nonholonomic vehicle models**, including higher-fidelity and three-dimensional systems

NASA Langley Research Center Hampton, VA
Research Intern, Guidance, Navigation, and Control (Analytical Mechanics Associates, Inc.) Jun. 2026 to Sept. 2026
Mentor: Dr. Luis G. Crespo

- Developing **model-free online optimization** for **formation-flight drag reduction**, enabling real-time performance-seeking without aerodynamic models
- Designing **online adaptive learning** and control architectures for autonomous **station-keeping** in wake-vortex flow fields, validated on a **high-fidelity fighter aircraft model** in closed-loop simulation
- Embedding **safety-critical filters** into the learning loop to guarantee safe operation during **online optimization**

Autonomous Systems Group, UT Austin Austin, TX
Undergraduate Research Assistant Jun. 2019 to Feb. 2020
Advisor: Prof. Ufuk Topcu

- Studied **multi-agent logistics and coordination** in confined environments such as warehouses
- Designed state-based controllers for **multi-quadcopter systems** using Slugs (reactive synthesis)
- Implemented and evaluated controllers in **AirSim** using custom **Unreal Engine** simulation environments using Python

OTHER EXPERIENCES AND PROJECTS

Instructional Assistant (UCSD) La Jolla, CA
Nonlinear Systems (MAE 281A) and Linear Control (MAE 142B) Jan. 2025 — Jun. 2025

- Awarded **MAE PhD Outstanding Teaching Assistant of the Year**, recognized for exceptional student support and instructional contributions.
- Led weekly discussion sections and office hours to clarify complex concepts and collaborated with faculty to refine course materials and provide consistent learning outcomes across sections.

NASA's 2020 RASC-AL Competition Finalist (Theme 5) Austin, TX
Project Autoponics - Team Lead Aug. 2019 — Jun. 2020

- Led the design and presentation of an autonomous plant habitat for the Lunar Gateway, **secured $11,000 funding** from the National Institute of Aerospace

Aerial Robotics Autonomy Protocol Project Austin, TX
Aerial Robotics Course Project Jan. 2020 — May 2020

- Developed a **C++ autonomy stack** for a quadcopter using 3D **A* path planning** and polynomial trajectory smoothing

Republic of Korea Air Force South Korea
Air Defense Artillery Brigade Oct. 2020 — Jul. 2022

- Led training of 20+ recruits in technical maintenance and tactical protocols

PUBLICATIONS

Journal Papers

[J1] **KH Kim**, M. Diagne and M. Krstić, *"Constant-Sum High-Order Barrier Functions for Safety Between Parallel Boundaries,"* in IEEE Control Systems Letters, vol. 9, pp. 1447-1452, 2025

Conference Papers

[C1] **KH Kim**, M Diagne, M Krstić, *"Robust Control Barrier Function Design for High Relative Degree Systems: Application to Unknown Moving Obstacle Collision Avoidance,"* in American Control Conference (ACC), Denver, CO, 2025

[C2] E. Zapien Ramos, **KH Kim**, M. Krstić, A. J. Rosengren, *"Safety-Critical Control Using Fully Nonlinear Equations of Relative Motion for Formation Flying in Cislunar Space,"* in AIAA SciTech Forum, 2026

[C3] **KH Kim**, V Todorovski, and M Krstić, *"Inverse Optimal Feedback and Gain Margins for Unicycle Stabilization,"* in American Control Conference (ACC), New Orleans, LA, 2026

[C4] V Todorovski, **KH Kim**, and M Krstić, *"Modular Design of Strict Control Lyapunov Functions for Global Stabilization of the Unicycle in Polar Coordinates,"* in American Control Conference (ACC), New Orleans, LA, 2026

[C5] M Krstić, V Todorovski, **KH Kim**, and A Astolfi, *"Integrator Forwarding Design for Unicycles with Constant and Actuated Velocity in Polar Coordinates,"* in American Control Conference (ACC), New Orleans, LA, 2026

[C6] M Krstić, **KH Kim**, and V Todorovski, *"Half-Global Deadbeat Parking for Dubins Vehicle,"* in American Control Conference (ACC), New Orleans, LA, 2026

[C7] **KH Kim**, V Todorovski, M Krstić, *"Global Exponential Stabilization of a 3D Nonholonomic Vehicle in Spherical Coordinates,"* accepted to Conference on Decision and Control (CDC) 2026.

[C8] V Todorovski, **KH Kim**, M Krstić, *"Global Exponential Stabilization of the Kinematic Bicycle Model of a Car in Polar Coordinates,"* accepted to Conference on Decision and Control (CDC) 2026.

Manuscripts in Review

[R1] V Todorovski, **KH Kim**, A Astolfi, and M Krstić, *"Nonholonomic Robot Parking by Feedback—Part I: Modular Strict CLF Designs,"* in review for IEEE Transactions on Automatic Control, Available: arXiv:2511.15119

[R2] **KH Kim**, V Todorovski, and M Krstić, *"Nonholonomic Robot Parking by Feedback—Part II: Nonmodular, Inverse Optimal, Adaptive, Prescribed/Fixed-Time and Safe Designs,"* in review for IEEE Transactions on Automatic Control, Available: arXiv:2511.15219

[R3] M Krstić, **KH Kim**, and V Todorovski, *"Dubins Vehicle Stabilization: Deadbeat Parking and Asymptotic 'Spinaway',"* in review for Automatica.

PROFESSIONAL EVENTS AND SERVICES

- Naval Innovation, Science, and Engineering Center (NISEC) Annual Project Review Talk (UCSD) May 2026
- Reviewer for IEEE Control Systems Letters (L-CSS), Conference on Decision and Control (CDC), and American Control Conference (ACC)
- RoboGrads Feed the Intellect (FTI) Seminar Talk Nov. 2024
- MAE Student Seminar Talk Nov. 2024

SKILLS

- **Autonomy & control:** Lyapunov-based methods (CLF/CBF), quadratic program (QP) safety filters, adaptive control, inverse-optimal control, PID, LQR, MPC
- **Learning-based methods:** Model-free online optimization (extremum seeking), Reinforcement learning methods (policy gradient, actor–critic, Q-learning, SARSA)
- **Signals & data analysis:** Filtering methods (low/high-pass), spectral analysis (DFT/FFT), system identification
- **Software:** MATLAB, Python, C++, Simulink, ROS 2, Git, SolidWorks, Microsoft Office
- **Languages:** English (native), Korean (native)

AWARDS

- MAE PhD Outstanding Teaching Assistant of the Year (UCSD) Jun. 2025
- Steve K. Sin Endowed Presidential Scholarship in Engineering (UT Austin) Jan. 2020
- University Honors (UT Austin) Aug. 2016 - May 2020